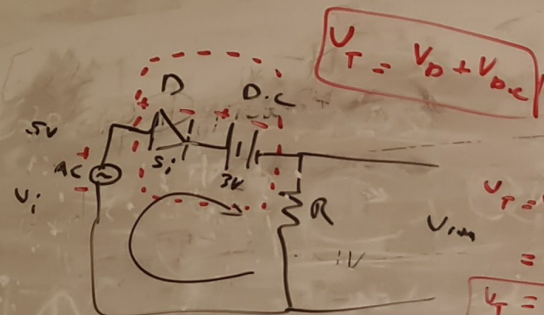


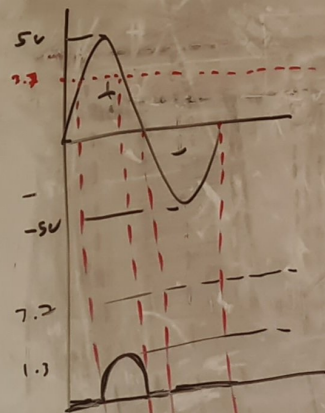


Clipper
with battery

$$V_T = V_D + V_{D.C.}$$

$$= 0.7 + 3$$

$$V_T = 3.7V$$



\$t + (1/2)\$ cycle

\$D\$ is off with A.C but \$R\$ with D.C

(\$I = 0\$, \$V_o = 0\$)

(Sh.C) A.C

\$V_i\$ (5V, 3.2)

\$V_i < V_T\$
\$D\$ is off

\$I = 0\$

$$V_o = IR = 0V$$

$$V_i \geq V_T$$

$$V_i = 5V$$

$$\text{by K.V.L}$$

$$\sum V_o = \sum V_i$$

$$V_o = V_i - V_T$$

$$= 5 - 3.7$$

$$V_o = 1.3V$$

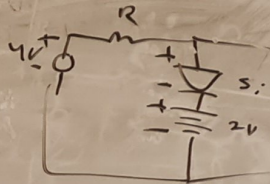
$$V_i = 3.7$$

by K.V.L

$$\sum V_o = \sum V_i$$

$$V_o = 3.7 - 3.7$$

$$V_o = 0V$$



$$V_T = V_D + V_{D.C.}$$

$$V_T = 2 + 0.7 = 2.7V$$

\$t + 1/2\$ cycle

\$D\$ is off with A.C but \$R\$ with D.C

$$V_i < V_T \text{ off}$$

$$V_o = V_i$$

$$V_i \geq V_T \text{ on}$$

$$V_o = V_T = 2.7V$$

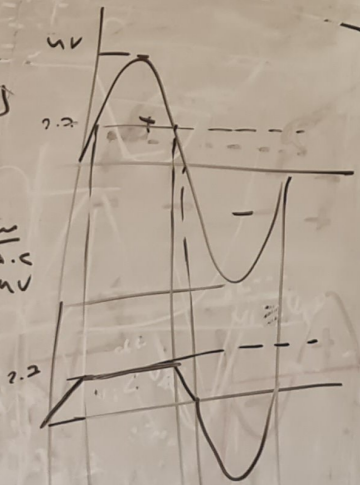
\$t + 1/2\$ cycle

\$D\$ is \$R\$ with A.C & D.C -4V

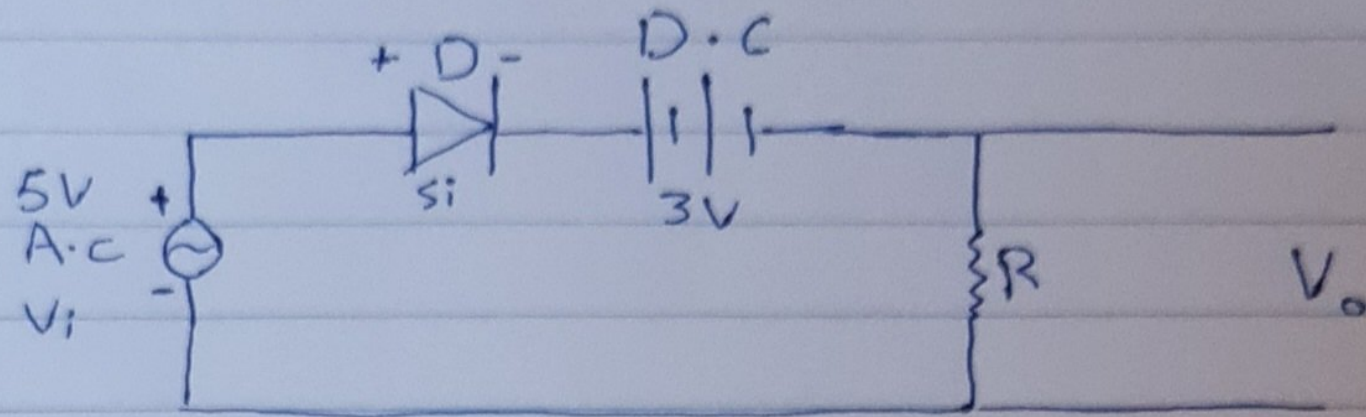
\$D\$ is off

$$V_o = V_i$$

$$V_o = -4V$$



Q 1.



What will happen if :

(A) $D.C = A.C$

(B) $D.C > A.C$ if $A.C = 3V$, $D.C = 5V$